

Terna Engineering College
Computer Engineering Department
Program: Sem VI

Course: Cloud Computing Lab (CSL603)

Faculty: Preeti patil

PART A

(PART A: TO BE COMPLETED BY STUDENTS)

Experiment No. 2

A.1 Aim:

Implement virtualization using Virtual box

A.2 Prerequisite:

Knowledge of Networking, Distributed Computing and knowledge of Software architectures.

A.3 Objective:

Understand different types of virtualizations, Host and bare metal hypervisors and create and run virtual machines

A.4 Outcome: (L02)

After successful completion of this experiment student will be able to

- understand the Virtualization
- Create Virtual machines using various hypervisors available

A.5 Theory:

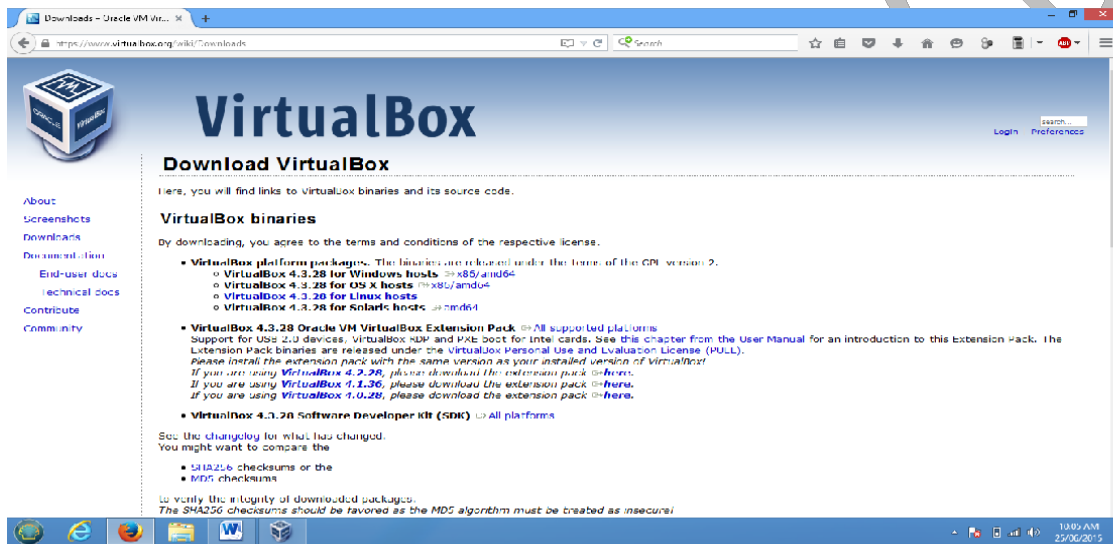
Steps: Creating and running virtual machines on Hosted Hypervisor like Virtual Box and KVM

2B

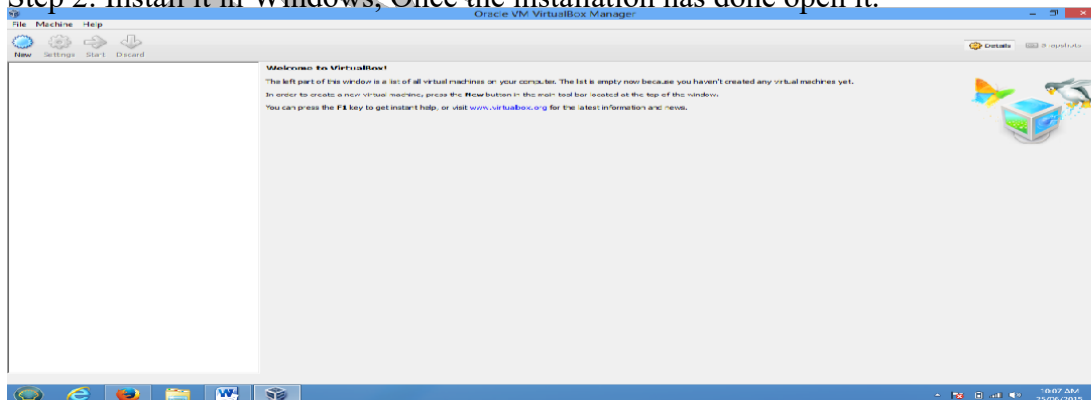
Creating and running virtual machines on Hosted Hypervisor like Virtual Box and KVM

1) Hosted Virtualization on Oracle Virtual Box Hypervisor

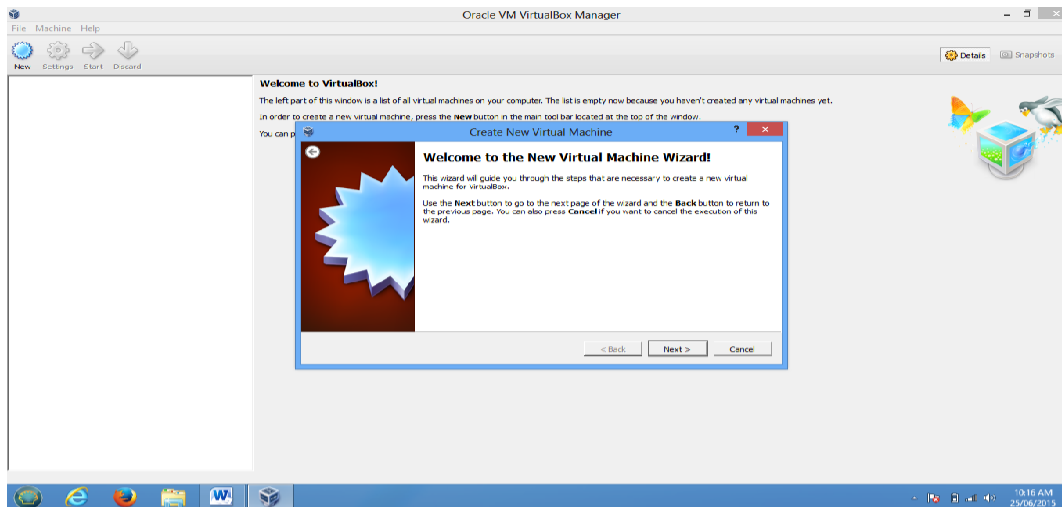
Step 1: Download Oracle Virtual box from <https://www.virtualbox.org/wiki/Downloads>



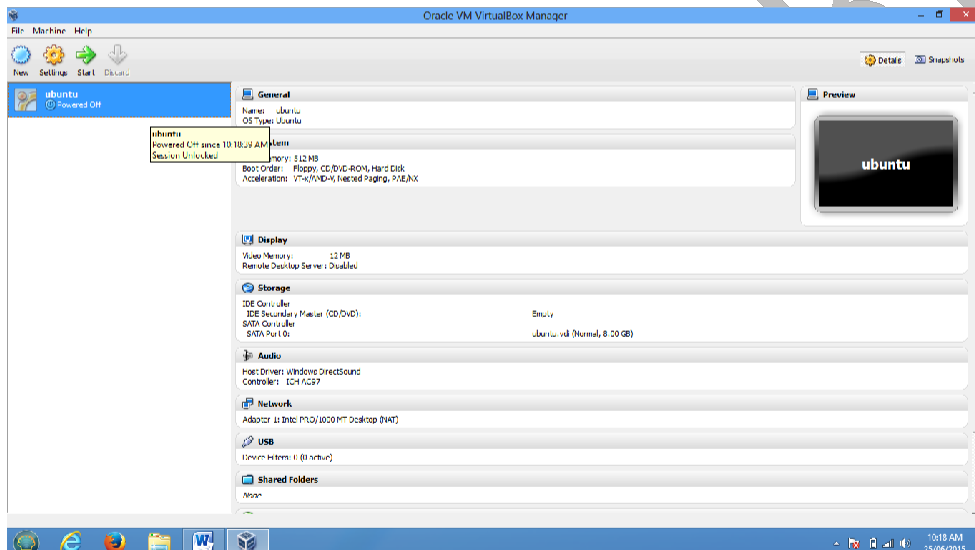
Step 2: Install it in Windows, Once the installation has done open it.



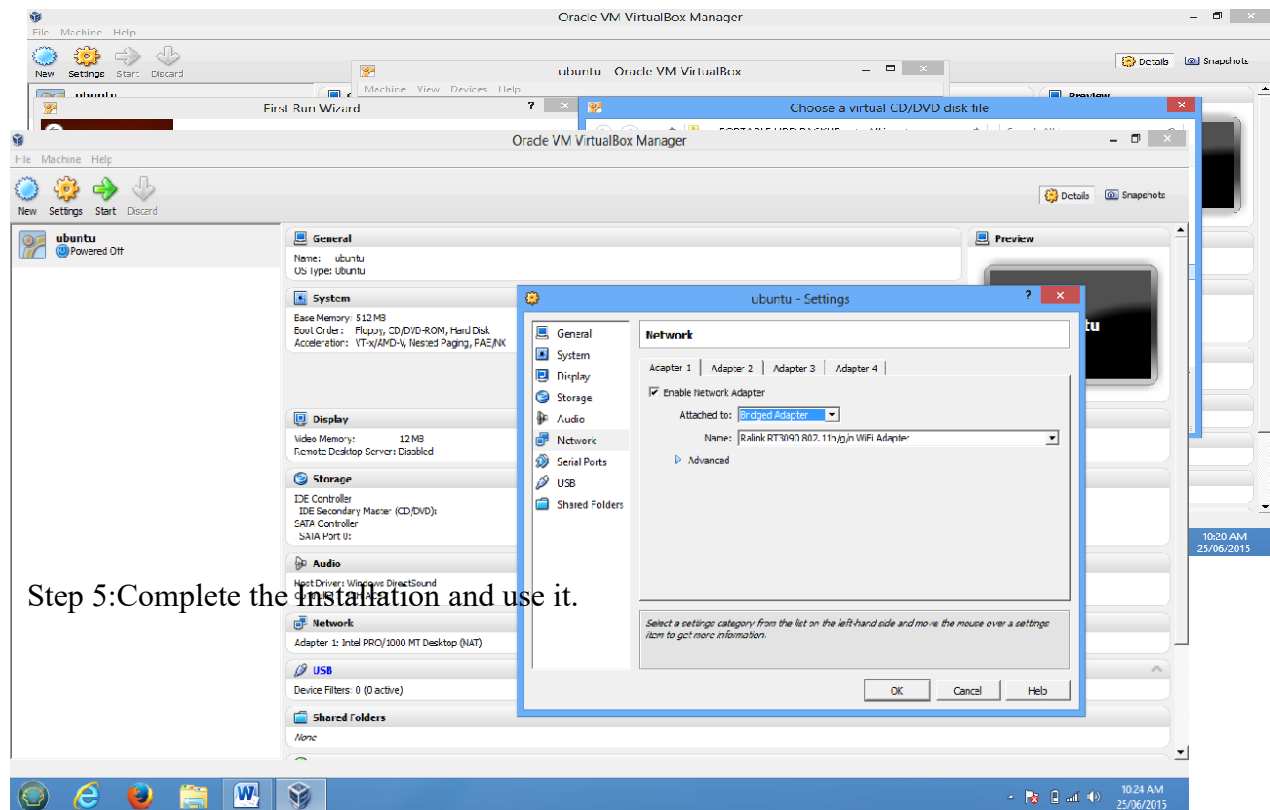
Step 3:- Create Virtual Machine by clicking on New



Step 4-: Specify RAM Size, HDD Size, and Network Configuration and Finish the wizard



Step 4-: To Select the media for installation Click on start and browse for iso file



```
root@ubuntu:/home/tsec# sudo grep -c "svm\|vmx" /proc/cpuinfo
3
root@ubuntu:/home/tsec#
```

Step 6: To Connect OS to the network change network Mode to Bridge Adapt

PART B

(PART B: TO BE COMPLETED BY STUDENTS)

(Students must submit the soft copy as per following segments within two hours of the practical. The soft copy must be uploaded on the ERP or emailed to the concerned lab in charge faculties at the end of the practical in case there is no ERP access available)

Roll No.: C-56	Name: Quazi Md Moinuddin
Class : TE Comps C	Batch : C-3
Date of Experiment: 17.01.25	Date of Submission: 5.02.25
Grade :	

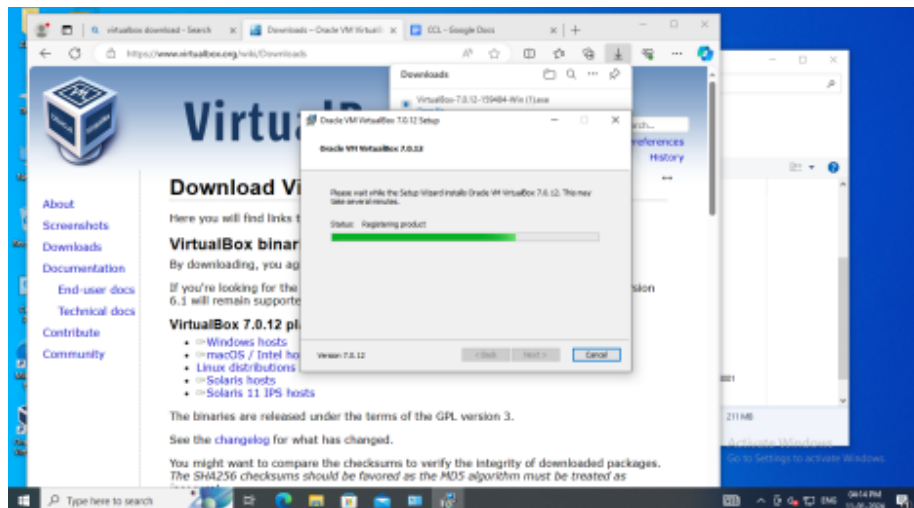
B.1 Question of Curiosity:

(To be answered by student based on the practical performed and learning/observations)

Q.1: Write down steps to install Virtual Box (with appropriate screen shot)..

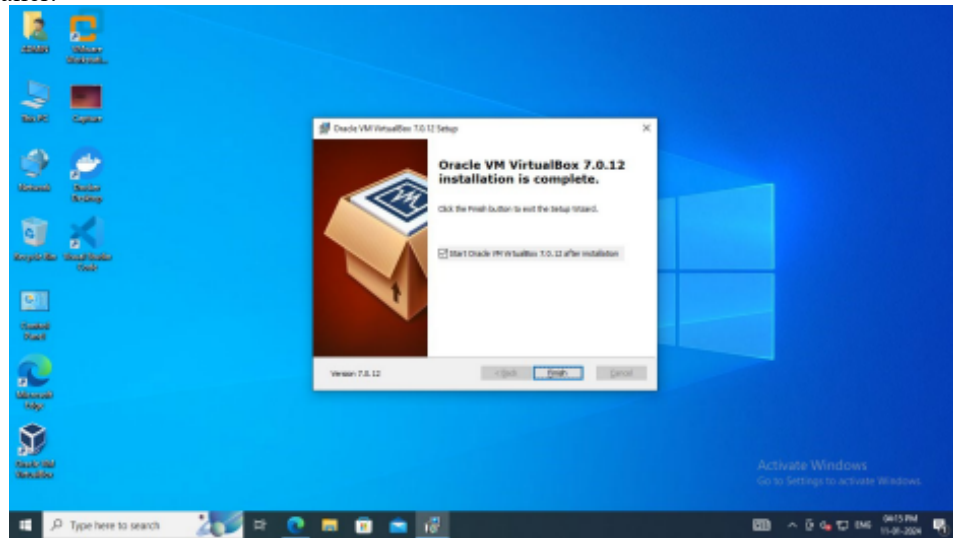
Installing VirtualBox:

- Download VirtualBox: Visit the official VirtualBox website at <https://www.virtualbox.org/> and navigate to the "Downloads" section.
- Select Your Host Platform: Choose the appropriate version for your operating system. For example, if you're using Windows, click on the "Windows hosts" link to download the installer.
- Download the Installer: Click on the download link to download the VirtualBox installer. Save the file to your computer.
- Run the Installer: Locate the downloaded installer file and double-click to run it. Follow the on-screen instructions to begin the installation process.



Installation Wizard: The installation wizard will guide you through the setup process. Click "Next" to proceed.

- Choose Components: Select the components you want to install. The default options are usually sufficient for most users. Click "Next" to continue.
- Choose Install Location: Specify the installation location or use the default. Click "Next" to proceed.
- Create Start Menu Entries: Choose whether you want to create shortcuts in the Start menu. Click "Next" to continue.
- Select Additional Tasks: Choose any additional tasks you want the installer to perform, such as creating desktop shortcuts. Click "Next" to proceed.
- Install: Review your choices and click "Install" to begin the installation process.
- Driver Installation (Windows Only): During the installation on Windows, you may be prompted to install device drivers. Allow the installation to proceed.
- Complete the Installation: Once the installation is complete, click "Finish" to exit the installer.



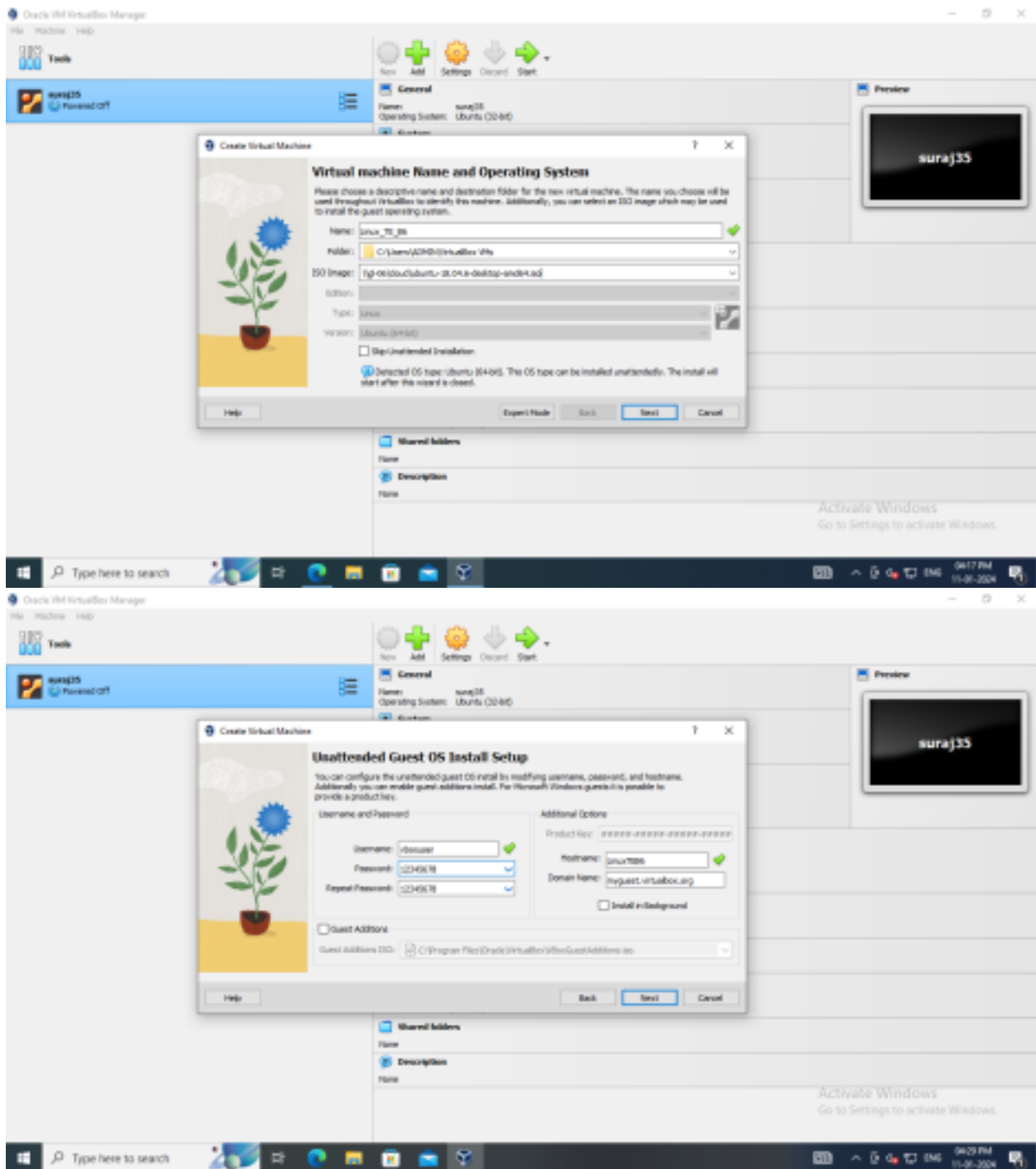
Q2: Write down steps to create virtual machines (with appropriate screen shot).

Creating a Virtual Machine in VirtualBox:

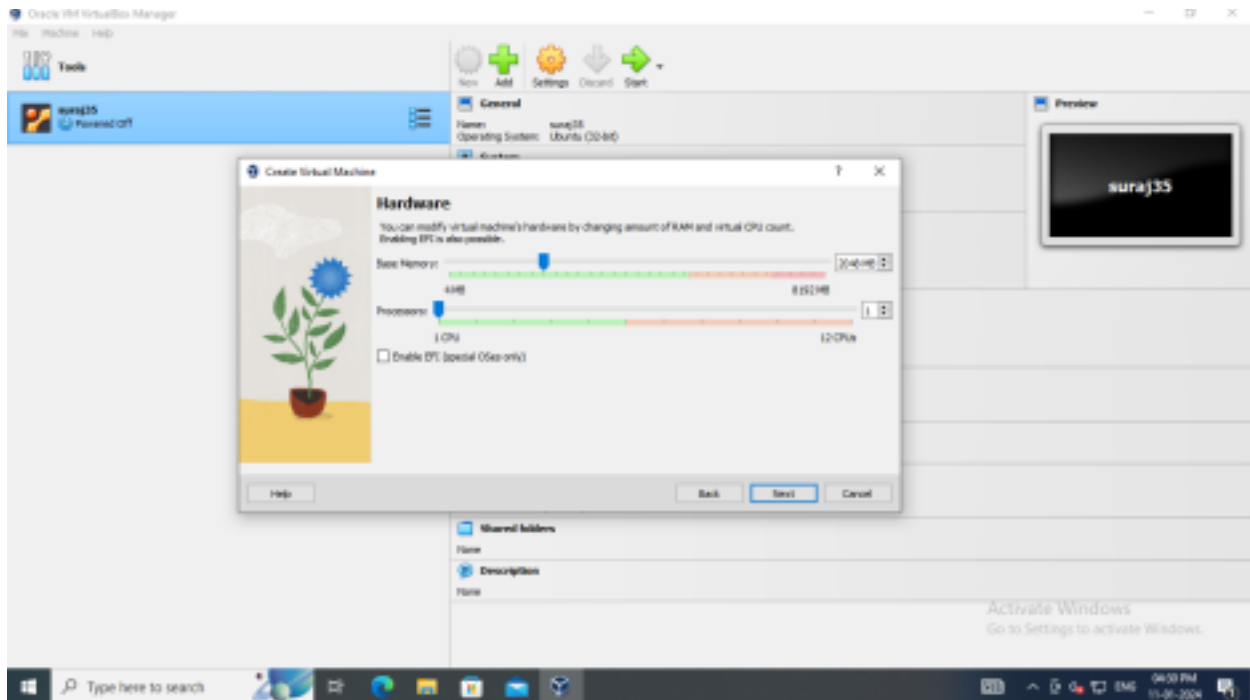
- Open VirtualBox: Launch the VirtualBox application.
- Click on "New": In the VirtualBox main window, click on the "New" button in the toolbar.
- Enter Name and OS Type:

Enter a name for your virtual machine.

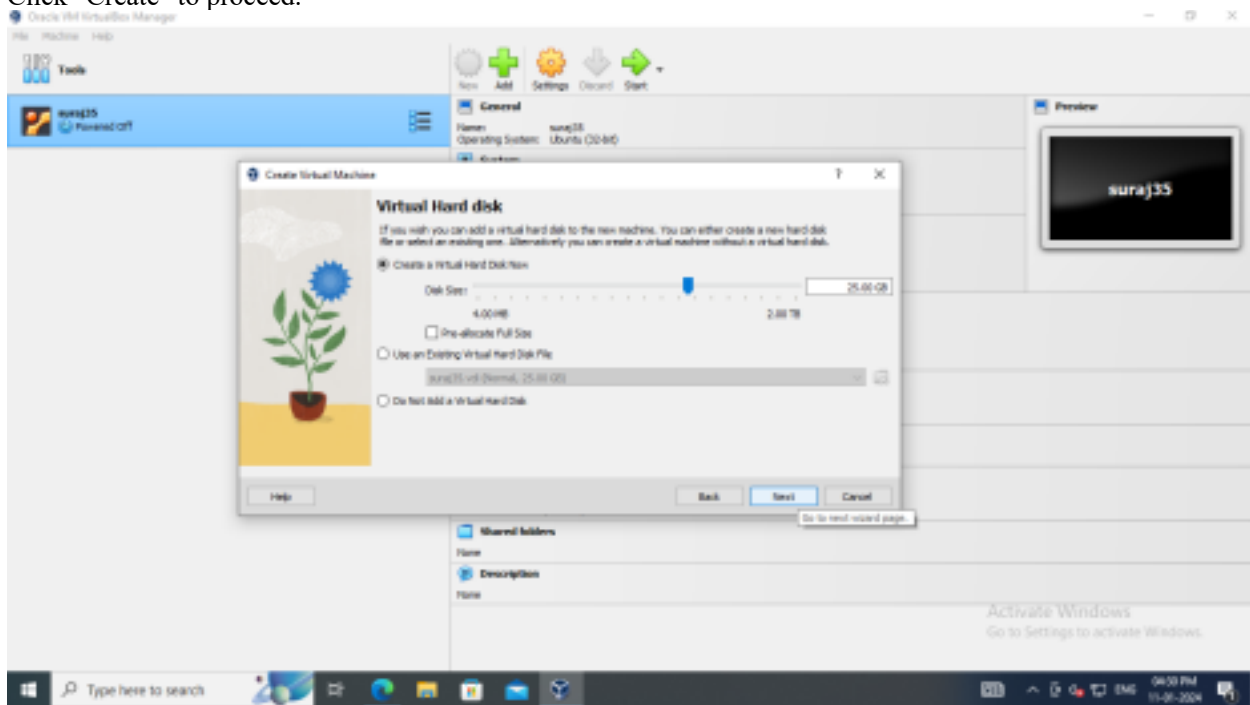
Choose the type of operating system (OS) and version. For example, if you're installing Windows, select "Microsoft Windows" and choose the version accordingly.



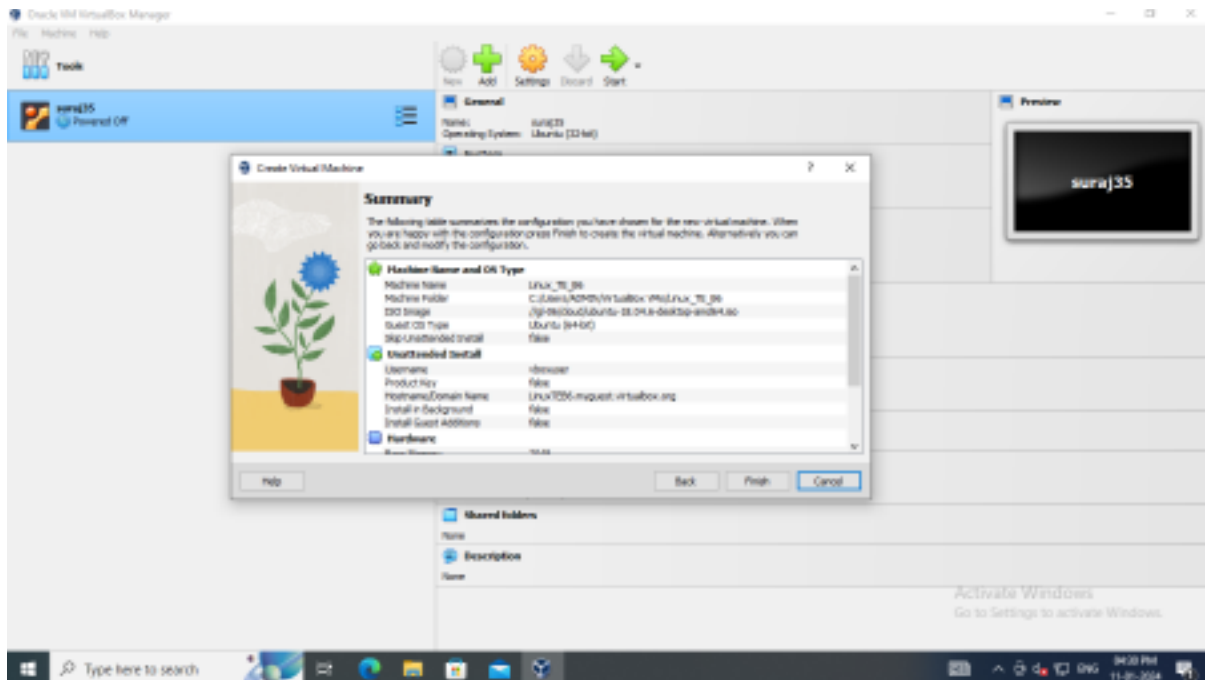
- Assign Memory (RAM):
Choose the amount of RAM to allocate to your virtual machine. Ensure it's within the green zone for optimal performance.



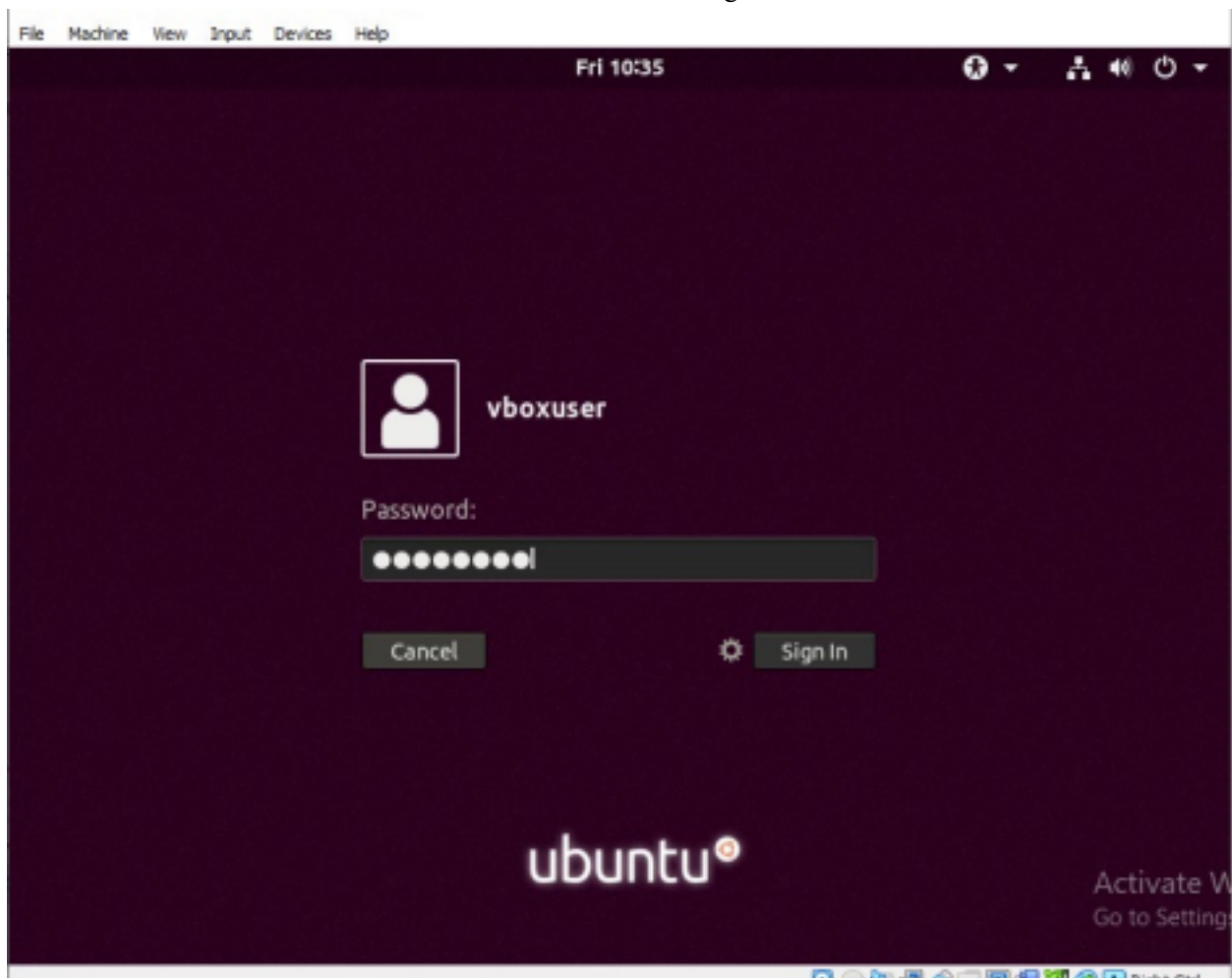
- Click "Next" to proceed.
- Create Virtual Hard Disk:
Choose to create a virtual hard disk now and click "Create."
Select the hard disk file type (usually VDI) and click "Next."
Storage on Physical Hard Disk:
Choose whether to allocate storage dynamically (expanding as needed) or fixed size (using a set amount of space).
Specify the size of the virtual hard disk.
Click "Create" to proceed.

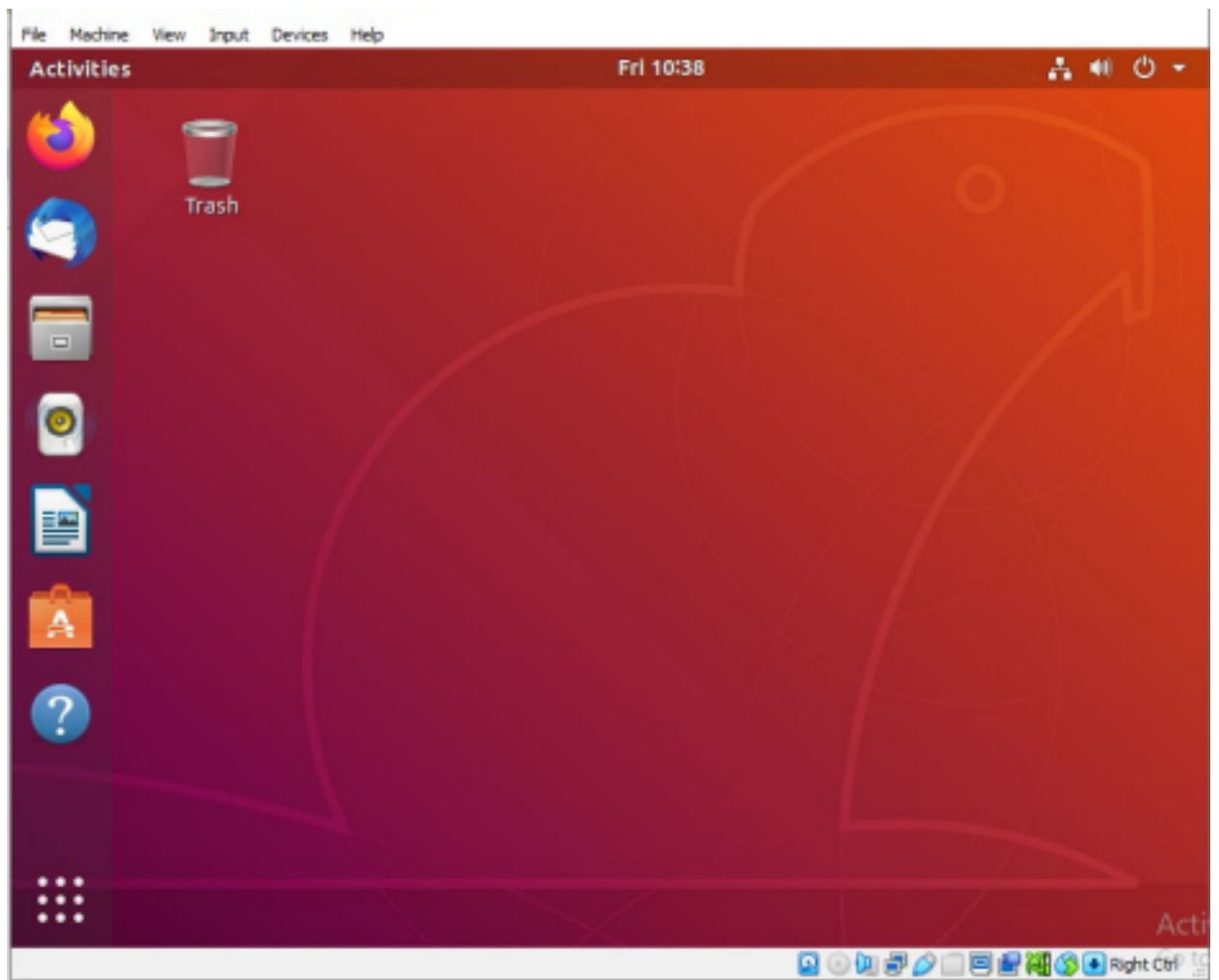


Finish Creating Virtual Machine:



You'll see the new virtual machine listed in the VirtualBox Manager.





Q3: Discuss various advantages of virtual machines.

- **Resource Utilization:**
 - **Server Consolidation:** Virtualization allows multiple virtual machines to run on a single physical server. This helps in consolidating workloads and making more efficient use of hardware resources.
 - **Optimized Resource Allocation:** VMs can be allocated specific amounts of CPU, memory, and storage, optimizing resource utilization based on the needs of individual workloads.
- **Isolation and Security:**
 - **Isolation:** Each VM operates independently of others on the same host, providing a level of isolation. This isolation helps prevent issues in one VM from affecting others.
 - **Security:** VMs can be encapsulated into files, making it easier to backup, restore, and secure the entire virtualized environment.
- **Flexibility and Portability:**
 - **Platform Independence:** VMs abstract the underlying hardware, making it possible to run VMs on different host systems without worrying about hardware compatibility.
 - **Migration:** VMs can be easily migrated or moved between different physical servers, enabling dynamic resource allocation and load balancing.
- **Hardware Independence:**
 - **Operating System Compatibility:** VMs can run different operating systems on the same physical hardware. This is beneficial for testing applications across various operating systems or running legacy applications on modern hardware.
- **Cost Savings:**
 - **Reduced Hardware Costs:** Through server consolidation, organizations can reduce the number of physical servers needed, leading to cost savings in terms of hardware, power, and cooling.
 - **Energy Efficiency:** Virtualization allows for better utilization of server resources, leading to improved energy efficiency and reduced operational costs.
- **Disaster Recovery and Redundancy:**
 - **Snapshot Backups:** VM snapshots, combined with backup solutions, facilitate efficient disaster recovery by providing quick restoration to a known state.
 - **High Availability:** Virtualization platforms often include features for high availability, ensuring that VMs are automatically moved or restarted on healthy hosts in the event of a hardware failure.
- **Testing and Development:**
 - **Isolated Test Environments:** VMs can be used to create isolated test environments, allowing developers and testers to work on applications without interfering with production systems.
 - **Rapid Prototyping:** VMs enable rapid prototyping and experimentation by quickly deploying new instances without the need for additional hardware.

- Legacy Application Support:
- VMs allow organizations to run legacy applications on modern infrastructure without compatibility issues. This helps in extending the lifespan of critical applications.

Q4: Differentiate between Xen, KVM, VMware hypervisors.

Xen:

- Type:
- Type 1 Hypervisor: Xen is a Type 1 hypervisor, meaning it runs directly on the bare-metal hardware, providing high performance and efficiency.
- Architecture:
- Microkernel Design: Xen employs a microkernel design, separating the hypervisor core from device drivers and other privileged components.
- Open Source:
- Open Source: Xen is an open-source hypervisor, widely used in both enterprise and cloud environments. It is included in many Linux distributions.
- Paravirtualization:
- Paravirtualization Support: Xen initially relied on paravirtualization, where guest operating systems are modified for better performance. However, it also supports hardware-assisted virtualization (HVM).
- Management Tools:
- XenServer: Citrix provides a commercial version of Xen called XenServer, which includes additional management tools and features.

KVM (Kernel-based Virtual Machine):

- Type:
- Type 1 Hypervisor: KVM is a Type 1 hypervisor that is integrated into the Linux kernel. It runs as a module within the Linux kernel and leverages the host operating system's scheduler.
- Architecture:
- Integrated with Linux Kernel: KVM is tightly integrated with the Linux kernel, making it a part of the host operating system.
- Open Source:
- Open Source: KVM is an open-source hypervisor and is part of the Linux mainline kernel. It benefits from regular updates and improvements contributed by the open-source community.
- Virtualization Support:
- Hardware-Assisted Virtualization: KVM primarily utilizes hardware-assisted virtualization (Intel VT and AMD-V) for improved performance.
- Management Tools:
- Libvirt: Libvirt is a common management API used for managing various

virtualization solutions, including KVM. Tools like virt-manager provide graphical user interfaces for managing KVM based virtual machines.

VMware:

- Type:
- Type 1 and Type 2 Hypervisors: VMware offers both Type 1 hypervisors (ESXi for bare-metal) and Type 2 hypervisors (Workstation, Fusion for desktops).
- Architecture:
- Proprietary Hypervisor: VMware's ESXi hypervisor has a proprietary architecture and is not based on the Linux kernel. It runs directly on the hardware for maximum performance.
- Virtualization Support:
- Hardware-Assisted Virtualization: VMware relies on hardware-assisted virtualization technologies (Intel VT and AMD-V) for efficient virtualization.
- Management Tools:
- vCenter: VMware provides vCenter Server for centralized management of multiple ESXi hosts. It offers advanced features like vMotion for live VM migration, High Availability (HA), and Distributed Resource Scheduler (DRS).
- Commercial Offerings:
- Commercial Solutions: VMware's hypervisors, especially ESXi, are widely used in enterprise environments. VMware offers a range of commercial solutions for virtualization, including vSphere for data center virtualization.
- Summary:
- Xen: Open-source Type 1 hypervisor with a microkernel design, suitable for both enterprise and cloud environments.
- KVM: Open-source Type 1 hypervisor tightly integrated with the Linux kernel, using hardware assisted virtualization for performance.
- VMware: Offers both Type 1 and Type 2 hypervisors with a proprietary architecture. Widely used in enterprise environments with advanced features and commercial offerings

B.2 Conclusion:

We studied different types of hypervisors their uses . We also conducted practical by installing Virtual Box and running Linux on it .